Self-supervised confidence-aware denoising imputation of genomic data

SFU

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The problem

Most (cell type, assay) experiments were never run, and those that were differ in depth, read length, run type and platform. In the ENCODE Imputation Challenge, a baseline that averages each assay over other cell types beat all but two submitted methods [1].

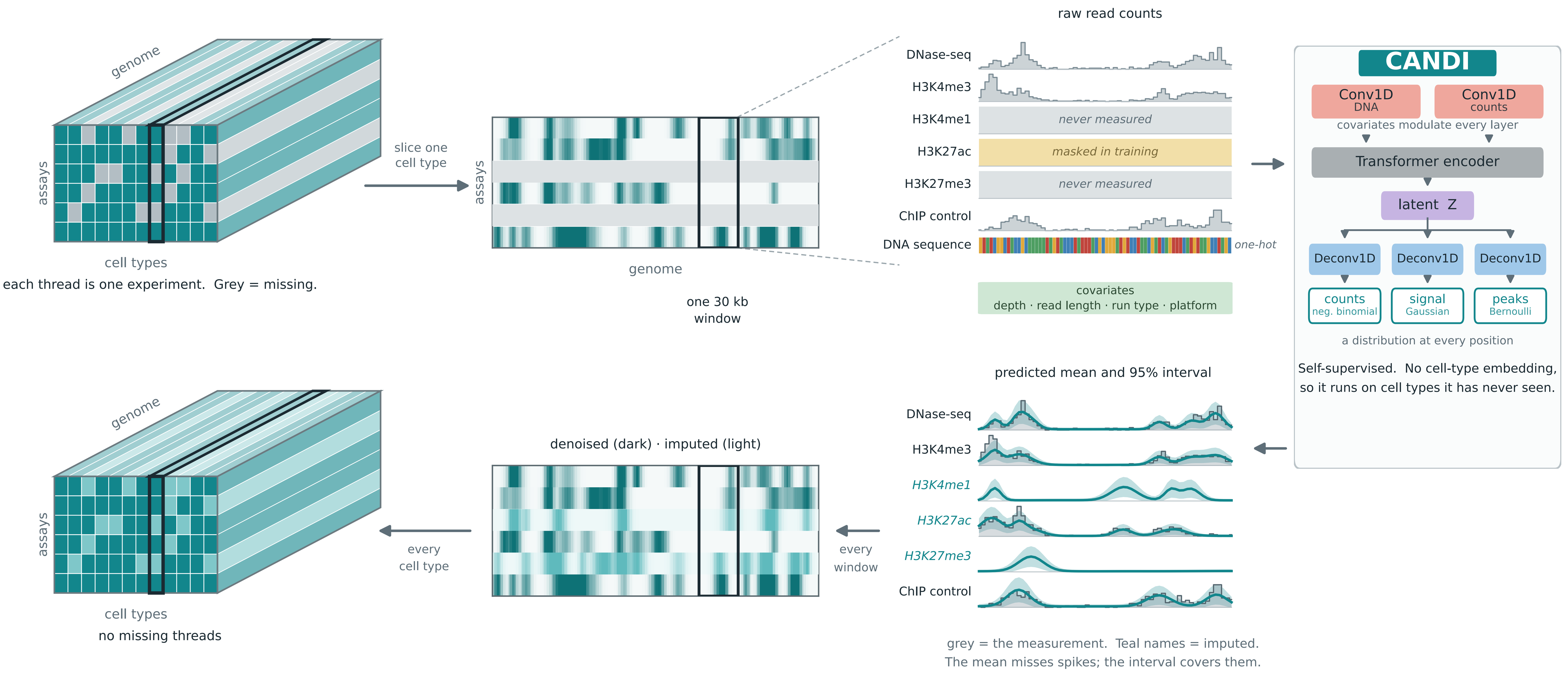
CANDI

Confidence-**A**ware **N**eural **D**enoising **I**mputer [2] is a self-supervised transformer over raw read counts, DNA sequence and four experimental covariates. It predicts a distribution for every assay in every 25 bp bin, so each imputed value has an interval. It has no cell-type embedding, so it runs on cell types it has never seen.

Take-home

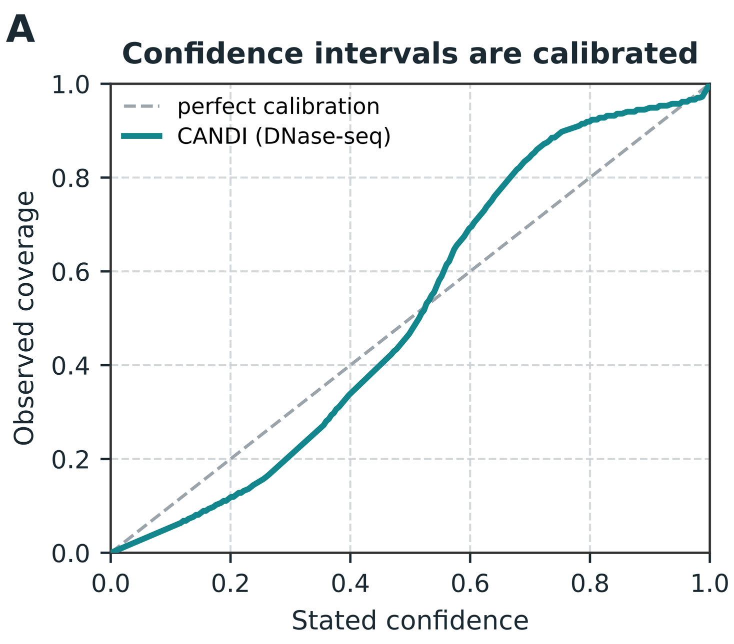
- CANDI is **1st of 26** on the ENCODE Imputation Challenge blind test.
- The lead needs a **24-number output correction**, fitted on training data only. Without it, CANDI is 14th.
- CANDI's intervals are calibrated, and its latent features give more reproducible chromatin states.

How CANDI works

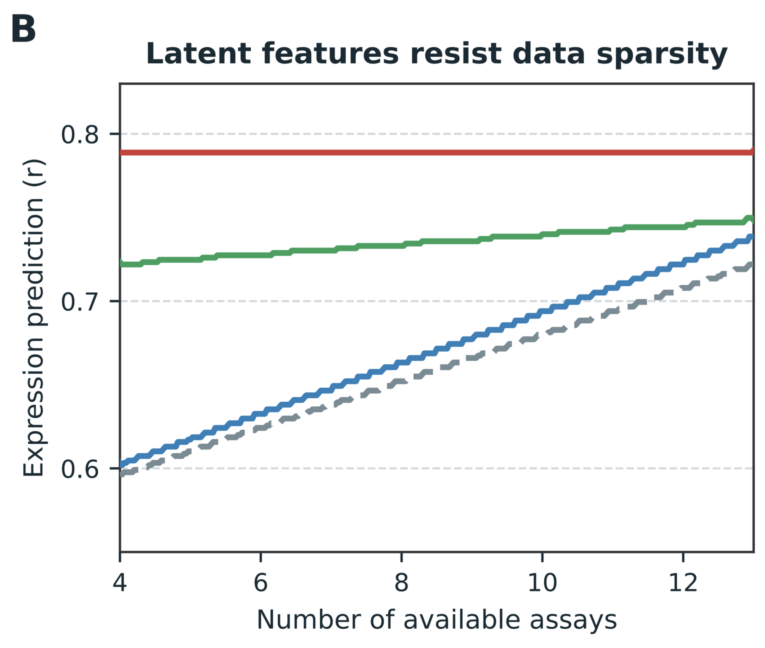

genome
assays
cell types
each thread is one experiment. Grey = missing.
slice one cell type
genome
one 30 kb window
raw read counts
DNase-seq
H3K4me3
H3K4me1
never measured
H3K27ac
masked in training
H3K27me3
never measured
ChIP control
DNA sequence
one-hot
covariates
depth · read length · run type · platform
CANDI
Conv1D DNA
Conv1D counts
covariates modulate every layer
Transformer encoder
latent Z
Deconv1D
Deconv1D
Deconv1D
counts neg. binomial
signal Gaussian
peaks Bernoulli
a distribution at every position
Self-supervised. No cell-type embedding, so it runs on cell types it has never seen.
predicted mean and 95% interval
DNase-seq
H3K4me3
H3K4me1
H3K27ac
H3K27me3
ChIP control
grey = the measurement. Teal names = imputed.
The mean misses spikes; the interval covers them.
every window
denoised (dark) · imputed (light)
every cell type
cell types
no missing threads

Top: one cell type is sliced out of the cell type × assay × genome tensor, and a 30 kb window of it goes into CANDI as raw counts, DNA and covariates. Grey assays were never measured; yellow ones are masked in training. **Bottom:** CANDI returns a mean and a 95% interval for every assay, and window by window this fills every missing thread of the tensor.

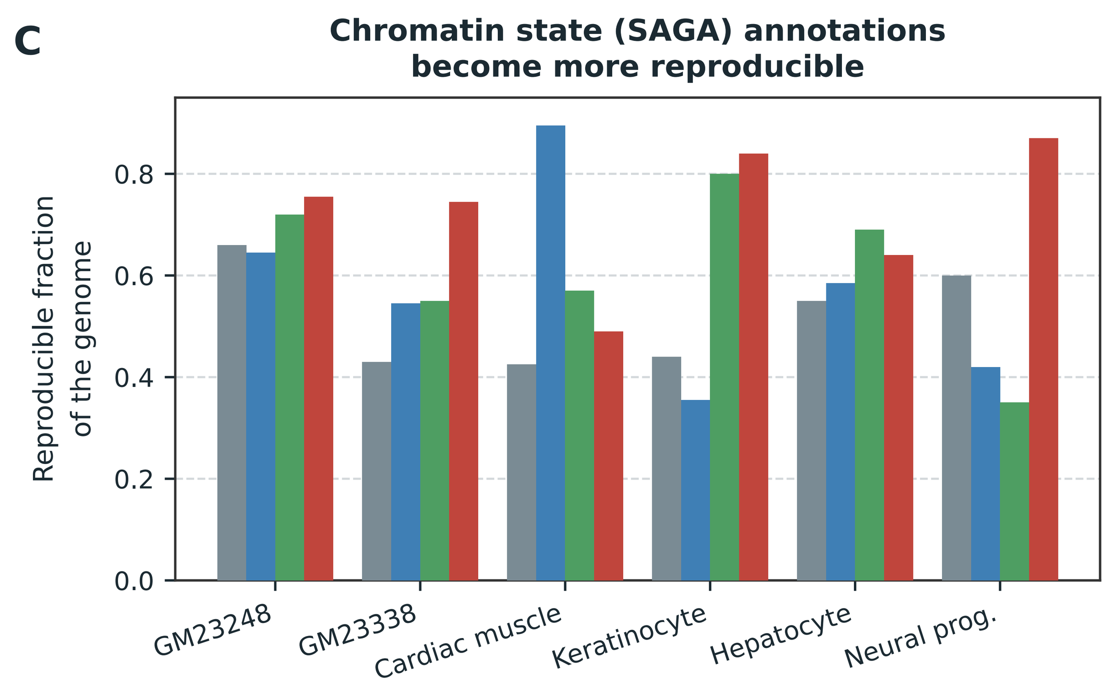
What the distributions and the latent features give you


A
Confidence intervals are calibrated
perfect calibration
CANDI (DNase-seq)
Observed coverage
Stated confidence
0.0
0.2
0.4
0.6
0.8
1.0

A For DNase-seq (cell line DND-41), the observed coverage of CANDI's intervals follows their stated confidence over the full range.


B
Latent features resist data sparsity
Expression prediction (r)
0.6
0.7
0.8
Number of available assays
4
6
8
10
12

B Gene expression predicted from each feature set as assays are removed. Latent features hold $r \approx 0.79$ from 13 assays down to 4; observed data falls to 0.60.

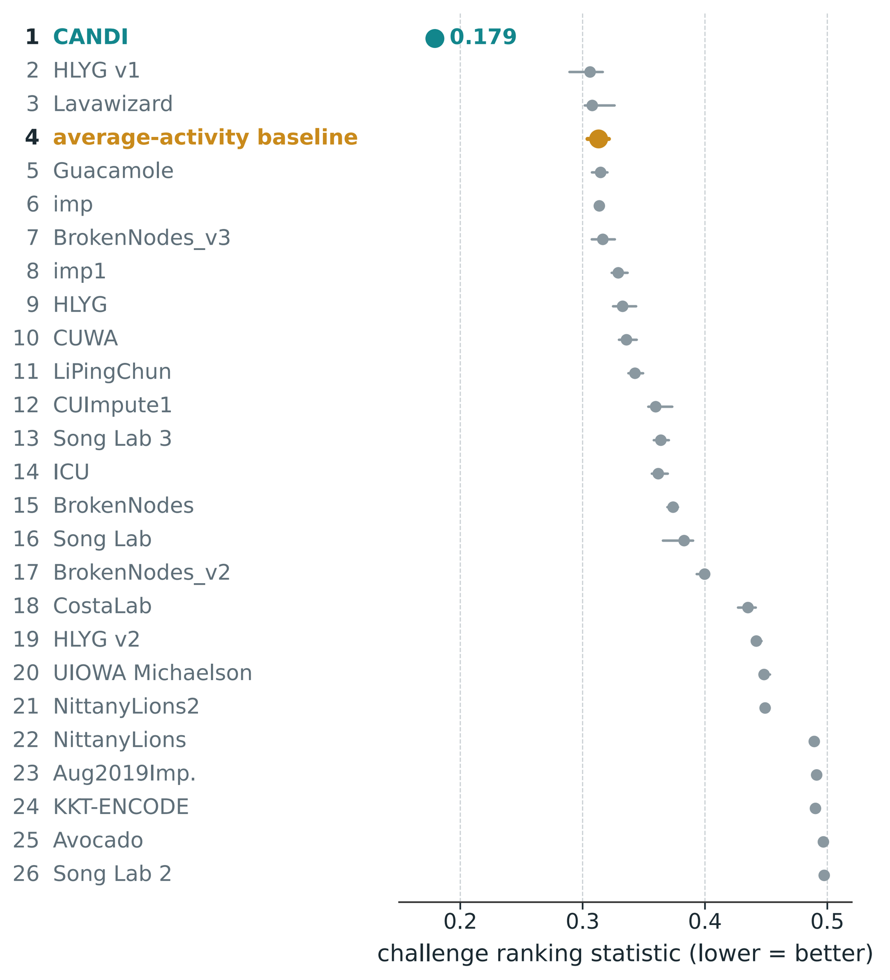

C
Chromatin state (SAGA) annotations become more reproducible
Reproducible fraction of the genome
0.0
0.2
0.4
0.6
0.8
GM23248
GM23338
Cardiac muscle
Keratinocyte
Hepatocyte
Neural prog.

C Agreement of SAGA chromatin-state annotations between replicates. Latent features give the most reproducible annotation in four of six biosamples.

Observed · Denoised · Denoised+Imputed · Latent

ENCODE Imputation Challenge: the blind test

51 blind-test experiments (12 cell types, 8 assays), scored with the challenge's own nine measures and ranking statistic. Our scorer reproduces every published entry to a median relative difference of 9×10^{-10}.


1 CANDI
0.179
2 HLYG v1
3 Lavawizard
4 average-activity baseline
5 Guacamole
6 imp
7 BrokenNodes_v3
8 imp1
9 HLYG
10 CUWA
11 LiPingChun
12 CUImpute1
13 Song Lab 3
14 ICU
15 BrokenNodes
16 Song Lab
17 BrokenNodes_v2
18 CostaLab
19 HLYG v2
20 UIOWA Michaelson
21 NittanyLions2
22 NittanyLions
23 Aug2019Imp.
24 KKT-ENCODE
25 Avocado
26 Song Lab 2
0.2
0.3
0.4
0.5
challenge ranking statistic (lower = better)

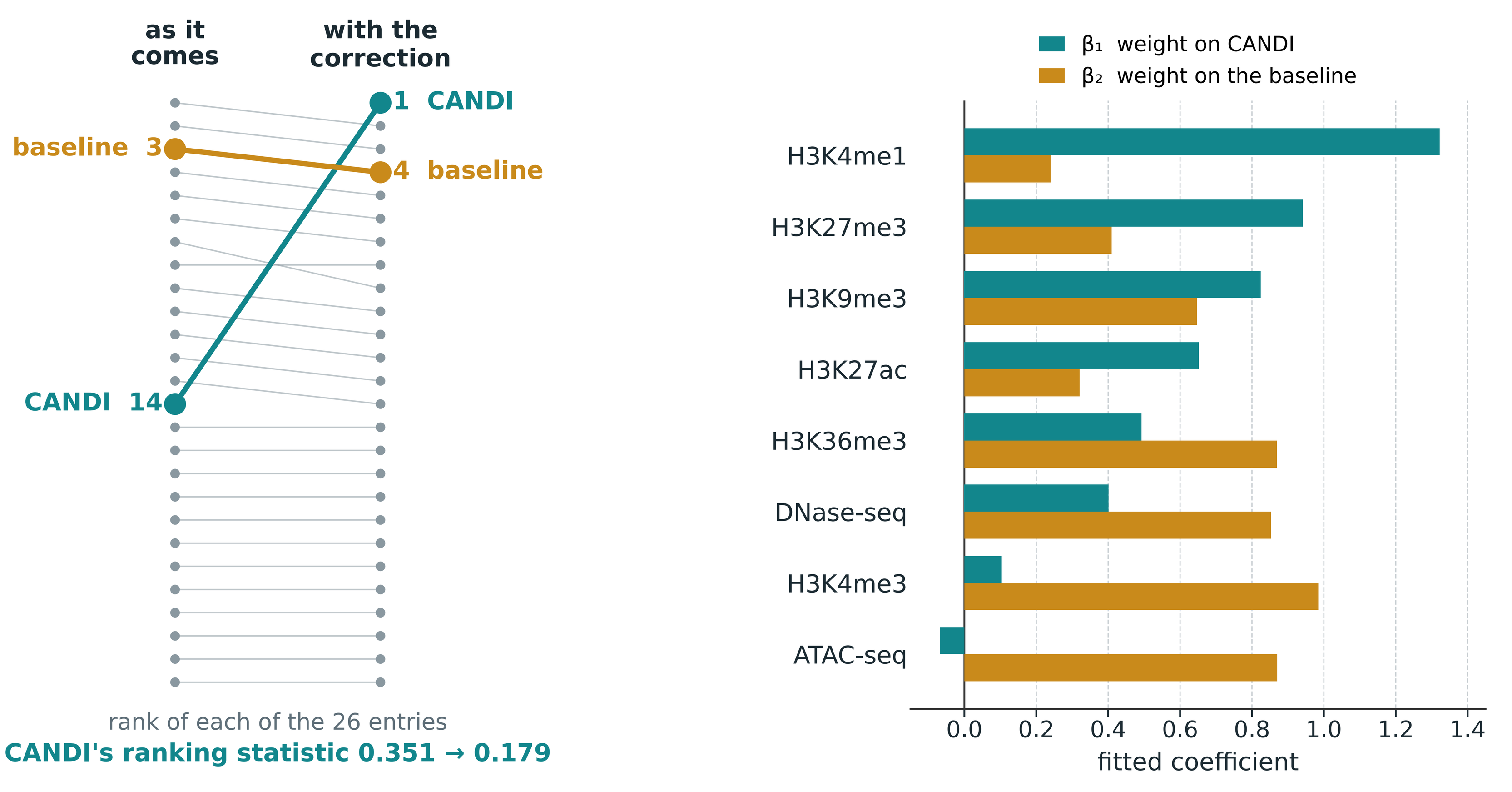
D The challenge's leaderboard with CANDI added. CANDI is first at 0.179 (bootstrap range 0.177–0.184); the next entry scores 0.306. Bars span the ten bootstrap groups.

What we changed in CANDI's output

For each assay, CANDI's track passes through one fixed affine layer:

$$\hat{y} \;=\; \max\big(\beta_0 \,+\, \beta_1 \cdot \text{CANDI} \,+\, \beta_2 \cdot \text{baseline},\; 0\big)$$

where *baseline* is the challenge's average-activity track for that assay. The 24 numbers (3 per assay × 8 assays) are fitted once, by least squares, on the 133 training experiments, where CANDI's predictions and the truth are both known. The model is not retrained, and no blind-test track is used.

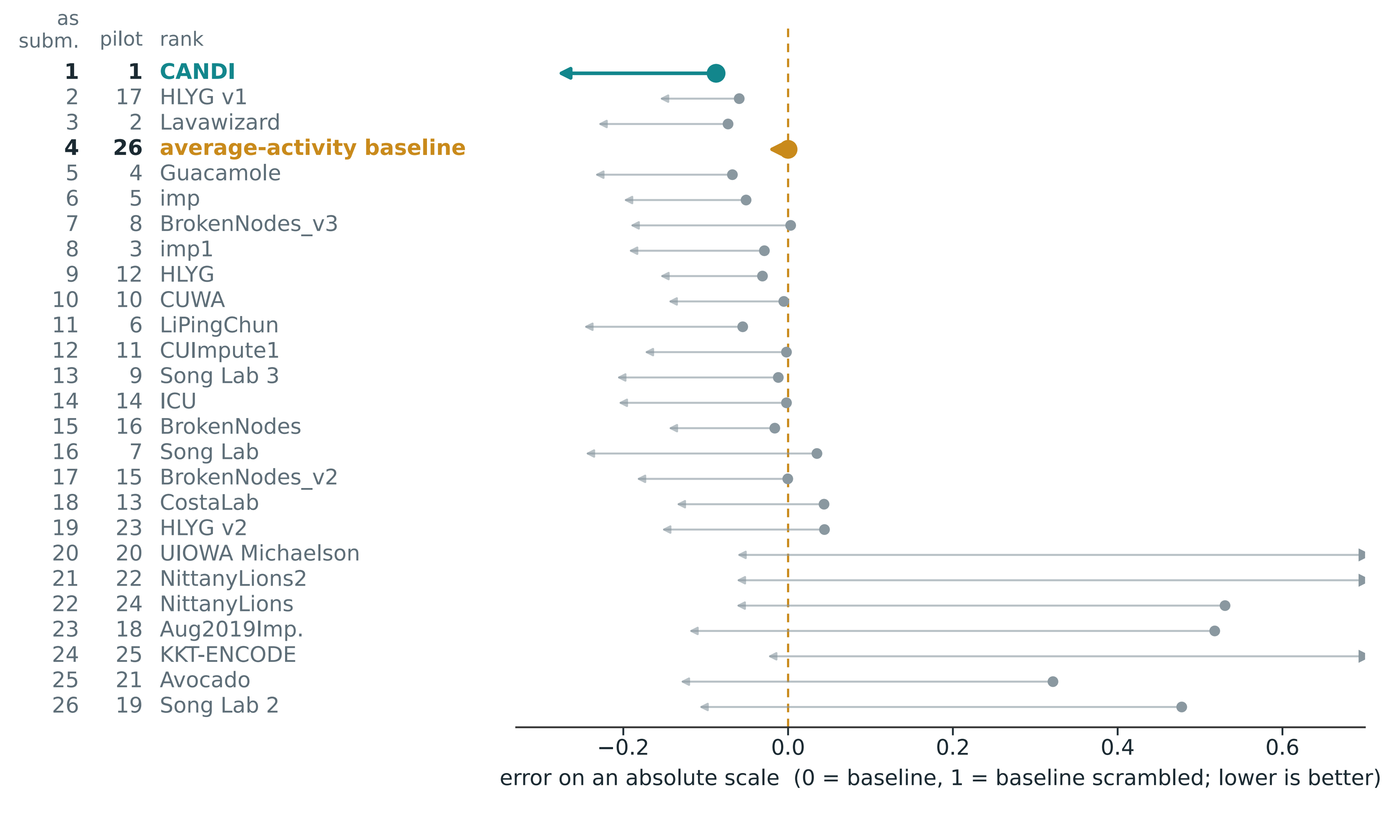

as it comes
with the correction
baseline 3
CANDI 14
1 CANDI
4 baseline
rank of each of the 26 entries
CANDI's ranking statistic 0.351 → 0.179
β₁ weight on CANDI
β₂ weight on the baseline
H3K4me1
H3K27me3
H3K9me3
H3K27ac
H3K36me3
DNase-seq
H3K4me3
ATAC-seq
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
fitted coefficient

E The correction moves CANDI from 14th to 1st. The fitted weights lean on CANDI for H3K4me1, H3K27me3 and H3K9me3, and on the baseline for H3K4me3 and the two accessibility assays.

We chose this correction after the challenge results were public. None of its numbers is fitted on blind-test data.


MSE, genome-wide
MSE, promoters
MSE, gene bodies
MSE, enhancers
MSE, variance-weighted
MSE, top 1% observed
MSE, top 1% imputed
Pearson r (reversed)
Spearman ρ (reversed)
better
CANDI
average-activity baseline
the 24 other entries

F Each of the nine measures relative to the baseline, median over the 51 experiments. CANDI is first on genome-wide MSE, Pearson and Spearman, and in the top 8 on five more.


as subm.
pilot
rank
1 1 CANDI
2 17 HLYG v1
3 2 Lavawizard
4 26 average-activity baseline
5 4 Guacamole
6 5 imp
7 8 BrokenNodes_v3
8 3 imp1
9 12 HLYG
10 10 CUWA
11 6 LiPingChun
12 11 CUImpute1
13 9 Song Lab 3
14 14 ICU
15 16 BrokenNodes
16 7 Song Lab
17 15 BrokenNodes_v2
18 13 CostaLab
19 23 HLYG v2
20 20 UIOWA Michaelson
21 22 NittanyLions2
22 24 NittanyLions
23 18 Aug2019Imp.
24 25 KKT-ENCODE
25 21 Avocado
26 19 Song Lab 2
−0.2
0.0
0.2
0.4
0.6
error on an absolute scale (0 = baseline, 1 = baseline scrambled; lower is better)

G Each entry again, after a rescale fitted on chr22 of its own target, which is not possible in practice. All 26 improve and CANDI stays first (0.243 against 0.266): much of the field's error is level, not shape.

mehdiforoozandeh.github.io/CANDI

[1] Schreiber J, Boix C, *et al.* The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

[2] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels D–G: blind-test scoring from the Libbrecht lab's EIC reproduction (experiments 025, 026, 029).